base value group here and propagate that assumption through uses of this lemma. In the cited proof, the essential step is that a nonempty definable subset of Γ contains a definable point; this fails for $(0,\infty)$ over the trivial base.

Author resolution. The author adopts nontrivial $\Gamma(\mathbb{S})$ as a standing assumption throughout the paper, matching the blanket hypothesis in [18]. It persists under parameter extension and supplies the required hypothesis here and in subsequent uses, including Lemma 2.25 and Corollary 2.30. The contrary permission at the opening of Section 6 is removed.

Lemma 2.24. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc. Then $\mathfrak{a}$ is t-atomic for all $t \in \mathrm{RV}$.*

Proof. If disc parameters have previously been named, the unary assertion follows directly from [18, Lemma 3.9], in its stated $\mathbb{U}^\bullet$ context: for any formula $\varphi(x,t)$ over this base, the definable map $x \mapsto \{s \in \mathrm{RV} : \varphi(x,s)\}$ is constant on the atomic open disc. Thus naming t does not split its type. For a real/Γ ground base the following argument also applies. For the case $n = 1$, if $\mathfrak{a}$ were not t-atomic then, by Corollary 2.21 and Lemma 2.13, it would contain a point algebraic over the fixed ground base. This point belongs to a finite ground-definable set. Atomicity would then force the whole infinite open disc $\mathfrak{a}$ to lie in that finite set, a contradiction. For the case $n > 1$, by induction on n, $\mathfrak{a}_1$ is t-atomic and for every $a \in \mathfrak{a}_1$, $\mathfrak{a}_{>1}$ is (a,t)-atomic. So $\mathfrak{a}$ cannot have a nonempty proper t-definable subset. □

Unapplied R029 proposal for author review. The new green sentence handles previously named disc parameters by the author-accepted unary result [18, Lemma 3.9]. It does not assert that Corollary 2.21 extends to arbitrary imaginary bases. The existing real/Γ proof and the product induction are retained. This proposal is not applied to the working manuscript.

Partly resolved referee R029 [Gap] – Atomicity proof: algebraic versus definable.
Original source line 886. Lemma 2.13 only gives an algebraic point over the original base, not necessarily a definable point. This suffices: an infinite atomic disc cannot meet a finite base-definable set. Replace 'definable point' at this step by that argument. The final sentence also needs 'nonempty proper t-definable subset'. Finally, indicate why Corollary 2.21 remains applicable whenever previously named disc parameters occur in the base; its stated hypothesis excludes such extra parameters.

Author resolution. The author approves both proof corrections: the point is algebraic over the fixed ground base, so atomicity would place the infinite disc inside a finite ground-definable set; and the final sentence now says "nonempty proper." The statement is unchanged. The separate question about applicability over additional disc-imaginary ground parameters remains open.

Lemma 2.25. *Suppose that $\Gamma(\mathbb{S})$ is nontrivial. Let $A \subseteq \mathrm{VF}^n$ be a set that generates a type and $\mathfrak{b} \subseteq A$ an open (or closed) polydisc. Then $\mathfrak{b}$ is $\ulcorner \mathfrak{b} \urcorner$-atomic.*

Proof. We do induction on n. The base case $n = 1$ is just [18, Lemma 3.4]. For the case $n > 1$, write $\mathfrak{b} = \prod_i \mathfrak{b}_i$, $\mathfrak{b}' = \prod_{i>1} \mathfrak{b}_i$, and $\gamma = \mathrm{rad}(\mathfrak{b}_1)$. For any $a \in \mathfrak{b}_1$, A_a generates a type. So, by Lemma 2.23 and the inductive hypothesis, $\mathfrak{b}'$ is $(a, \gamma, \ulcorner \mathfrak{b}' \urcorner)$-atomic and hence $\ulcorner \mathfrak{b} \urcorner$-atomic. Similarly $\mathfrak{b}_1$ is $\ulcorner \mathfrak{b} \urcorner$-atomic. It follows that $\mathfrak{b}$ must be $\ulcorner \mathfrak{b} \urcorner$-atomic. □

Lemma 2.26. *Let $\mathfrak{a} \subseteq \mathrm{VF}^n$ be an atomic open polydisc and $f : \mathfrak{a} \longrightarrow \mathcal{P}(\mathrm{VF})$ a definable finitary function such that $\mathrm{rv} \restriction f(a)$ is injective for every $a \in \mathfrak{a}$. Then $\mathrm{rv} \circ f$ is constant.*

Proof. This is clear since, by Lemma 2.24, for every $t \in \mathrm{rv}(\bigcup f(\mathfrak{a}))$, $\mathfrak{a}$ is t-atomic and hence f induces a t-definable function $\mathfrak{a} \longrightarrow t^\sharp$. The lemma follows. □

A *thin annulus* is a set of the form $\mathfrak{c}(a,\gamma) \setminus \mathfrak{o}(a,\gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x-a) = \gamma\}$, where $a \in \mathrm{VF}$ and $\gamma \in \Gamma$. Equivalently, it is a closed disc with one maximal open subdisc removed.

Lemma 2.27 ([18, Lemma 3.8])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc or an atomic closed disc or an atomic thin annulus. Let $f : A \longrightarrow \mathrm{VF}$ be a definable function. Then $f(A)$ is also atomic of one of these three forms.*

Resolved referee R163 [Clarification] – Define thin annuli before using them.
Original source line 907. The term "thin annulus" has not been defined in this manuscript. Specify a closed disc with one maximal open subdisc removed; after choosing the center of the removed subdisc, it is a shell $\{x : \mathrm{val}(x-a) = \gamma\}$. This makes the alternatives here and the later component argument unambiguous.

Author resolution. The author approves the finite-radius shell definition $\mathfrak{c}(a,\gamma) \setminus \mathfrak{o}(a,\gamma) = \{x \in \mathrm{VF} : \mathrm{val}(x-a) = \gamma\}$, with $\gamma \in \Gamma$. It is now stated before the first use in Lemma 2.27.

Lemma 2.28 ([18, Lemma 3.13])**.** *Let $A \subseteq \mathrm{VF}$ be an atomic open disc and $f : A \longrightarrow \mathrm{VF}$ a definable function. Then either f is constant or $f(A)$ is also an atomic open disc.*

Lemma 2.29. *Let $\mathfrak{a}$, f be as in Lemma 2.26 and suppose that f is not constant. Then there are $t_1, \ldots, t_k \in \mathrm{RV}_\infty$ and for each i, a t_i-definable function $f_i : \mathfrak{a} \longrightarrow t_i^\sharp$ such that $f = \bigcup_i f_i$ and each $f_i(\mathfrak{a})$ is either a point or a t_i-atomic open disc.*

Proof. By Lemma 2.26 and injectivity of rv on each finite fiber, f decomposes into finitely many t-definable single-valued branches $g : \mathfrak{a} \longrightarrow t^\sharp$. A branch with $t = \infty$ is the constant zero branch. Any constant branch already has the required form, so fix a nonconstant branch, for which $t \in \mathrm{RV}$. Lemma 2.24 makes $\mathfrak{a}$ atomic over t.

Every coordinate line of $\mathfrak{a}$ is atomic over its fixed coordinates and t. By Lemma 2.28, its image under g is a point or an open disc. Any two points of $\mathfrak{a}$ can be joined by changing one coordinate at a time. Starting at a point with image p and ending at a point with different image, the first nonconstant line image is an open neighborhood of p contained in $g(\mathfrak{a})$. Thus $g(\mathfrak{a})$ is open. It is a proper definable subset of VF, since it lies in $t^\sharp$.

By C-minimality, every point of a proper definable open subset of VF lies in a maximal open disc contained in that subset. Choose such a disc $D \subseteq g(\mathfrak{a})$ containing one value of g. An open line image that meets D is contained in D: intersecting open discs are nested, and maximality excludes a larger one inside $g(\mathfrak{a})$. A point image also stays in D when its initial value is in D. The coordinate chains therefore keep every value of g in D, so $g(\mathfrak{a}) = D$. The definable image of an atomic set is atomic over the same base, hence this open disc is t-atomic, as required.

□

Resolved referee R030 [Gap; revised] – A unary image lemma is used on a polydisc.
Original source line 918. For $n > 1$, Lemma 2.27 does not directly apply to $f : \mathfrak{a} \longrightarrow \mathrm{VF}$, since it has a unary domain. Supply the missing argument that $f(\mathfrak{a})$ has just one primitive component. A possible route is the canonical finite component decomposition of an atomic unary set: every coordinate slice of $\mathfrak{a}$ is atomic over its fixed coordinates, and its image is a point or an open disc by induction. Show that such an image must lie in one primitive component, then connect any two source points by changing coordinates one at a time. The same coordinate-chain argument excludes a closed disc or thin annulus as the whole image, because each slice image lies in a single maximal open subdisc. This fills the role of the inapplicable unary citation, once the component claim is spelled out.

Author resolution. The author approves the coordinate-chain proof. Unary coordinate-line images are points or open discs; maximal open discs in a proper definable open image then force the entire nonconstant image to be one open disc. This replaces the inapplicable unary citation on the polydisc. The tag ranges here and in Corollary 2.30 are also enlarged to RV_∞, retaining the constant zero branch.